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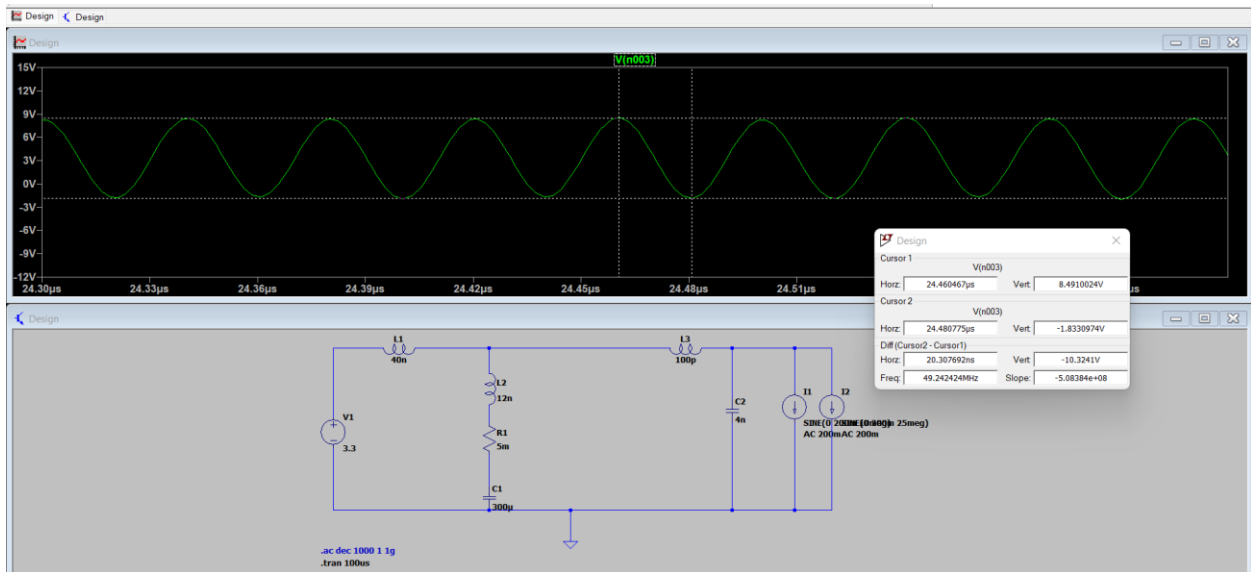


Fig 1: Voltage ripple without adding filter capacitor

From that figure we can see that the ripple voltage is high which will create noise to the circuit. So to decrease the noise we have to add filter capacitor to our circuit

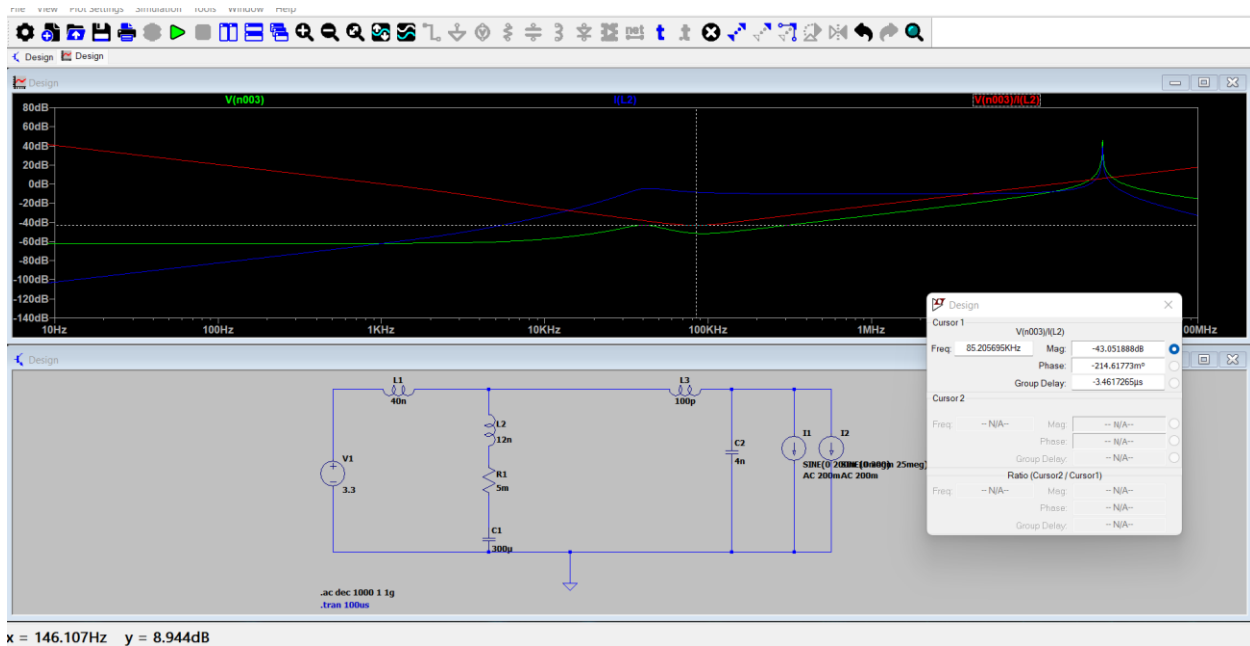


Fig 2: Impedance at resonant freq.

We can see that at the resonant frequency the impedance is low as at resonant frequency $L=C$ and the impedance becomes R only. so we just get the lowest frequency. Another thing is that the circuit impedance and the red line shows the impedance of a simple PDN seen from the load pin of the ic.

Now I will examine the simulation result with filter capacitor:

Given,

Two current :-
 DC current = 0
 AC current = 200 mA

Supply voltage $V_{pp} = 10 \text{ mV}$
 $ESR = 10 \text{ m}\Omega$
 $ESL = 4 \text{ nH}$

Task:- choose two capacitor's (< 100 each) &
 (i) calculate ripple reduction, and
 (ii) AC & transient simulations

Soln:

For 25 MHz ,

$$C = \frac{1}{(2\pi f)^2 \times L}$$

$$= \frac{1}{(2\pi \times 25 \times 10^6)^2 \times 4 \times 10^{-9}}$$

$$= 10.13 \times 10^{-9}$$

$$\approx 10 \text{ nF}$$

For, 10 MHz ,

$$C = \frac{1}{(2\pi \times 10 \times 10^6)^2 \times 4 \times 10^{-9}}$$

$$= 63.32 \times 10^{-9}$$

$$\approx 63 \text{ nF}$$

SHOT ON MI 9T
 AI TRIPLE CAMERA

Fig:3 Hand calculation of finding the value for Capacitor's value

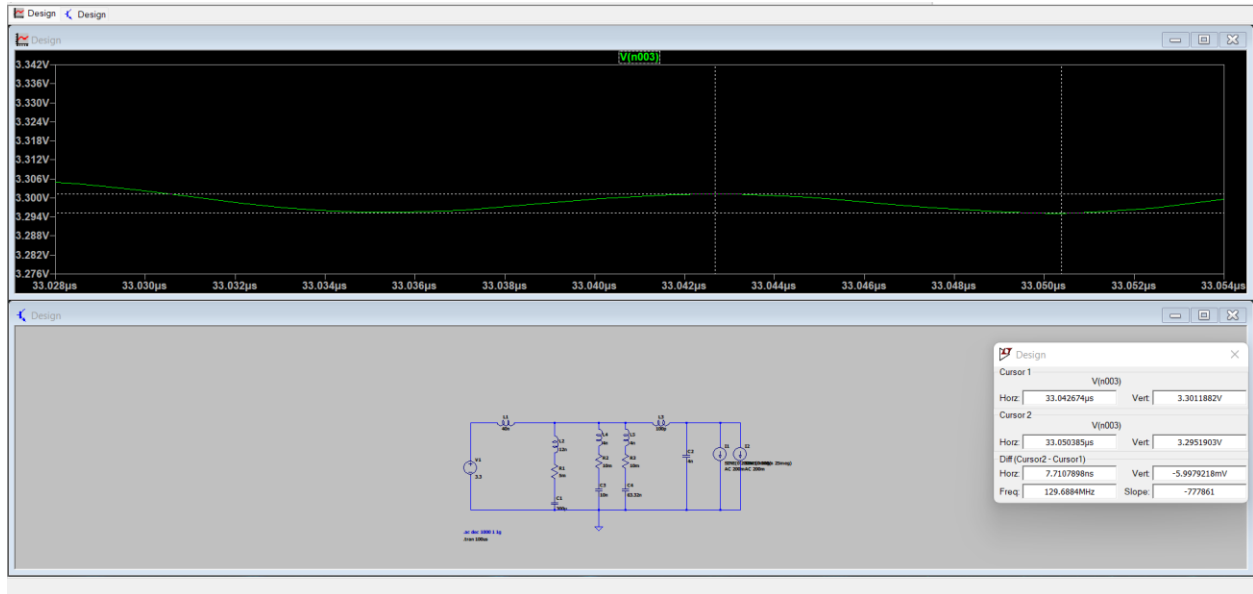


Fig.4: Voltage ripple after using filter capacitor.

We can clearly see that the voltage ripple decreases below 10mV and it becomes -5.99mV because of using the filter capacitance as decoupling capacitor.

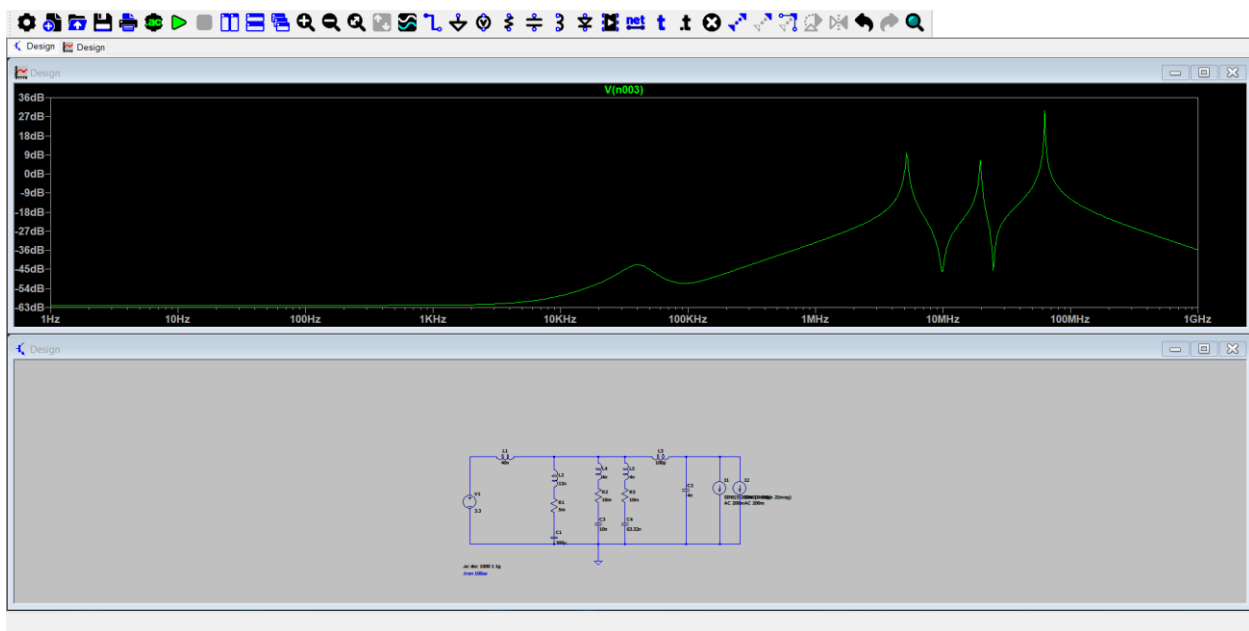


Fig. 5: Impedance after using filter capacitor.

After adding the filter capacitor, the power distribution network's impedance response is dramatically improved. The capacitor successfully filters out the high-frequency noise, which significantly reduces the peak-to-peak voltage ripple to an acceptable level. Without the capacitor, the impedance is high at the noise frequencies, leading to a much larger voltage ripple that can cause system instability.

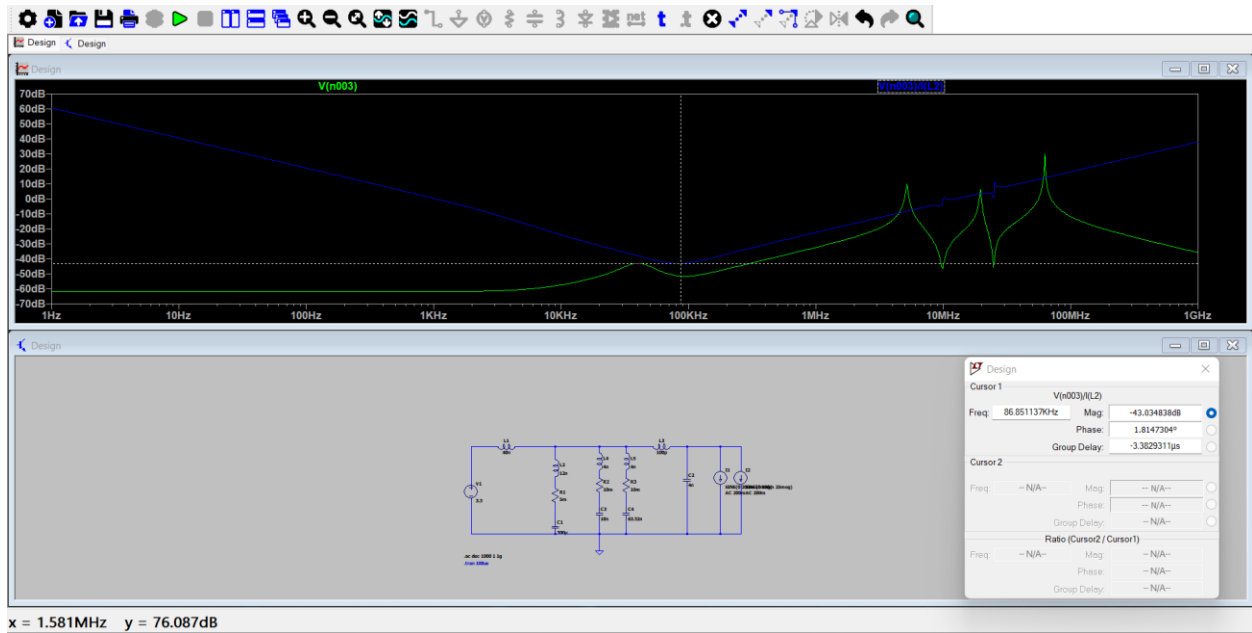


Fig.6: The resonant frequency impedance curve after using the filtering capacitor